

COMBIENBAR, Australia

WMO: 949140

Lat: 37.3417S

Lon: 149.0228E

Elev: 2103

StdP: 13.61

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.1	37.5	21.6	17.5	51.8	26.0	21.6	49.4	16.7	46.9	14.6	46.9	5.0	230	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.3	90.1	65.3	85.8	64.3	82.1	63.6	69.1	82.1	67.4	79.9	65.8	77.6	4.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	100.8	73.8	63.4	94.6	71.6	61.7	89.0	70.2	34.5	82.2	33.0	79.7	31.7	77.5	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	14.2	11.7	9.6	33.3	97.6	1.1	4.6	32.5	100.9	31.9	103.6	31.3	106.2	30.4	109.5	
(5)				31.6	72.7	0.9	1.5	30.9	73.8	30.4	74.7	29.9	75.6	29.2	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.6	66.1	65.0	61.6	56.1	51.4	47.4	46.2	47.5	51.0	54.6	58.7	61.9	(6)
(7)		DBStd	9.48	8.56	7.77	7.24	6.08	5.43	4.26	4.03	4.96	6.09	7.35	8.09	8.27	(7)
(8)		HDD50	507	0	0	2	13	46	100	131	110	58	33	11	3	(8)
(9)		HDD65	3829	91	92	157	274	422	527	583	543	421	333	225	161	(9)
(10)		CDD50	2541	499	419	361	196	90	23	13	32	88	175	272	373	(10)
(11)		CDD65	387	124	90	52	7	0	0	0	0	1	11	36	66	(11)
(12)		CDH74	3653	1307	793	367	33	0	0	0	0	8	98	342	705	(12)
(13)		CDH80	1282	546	267	100	3	0	0	0	0	1	16	97	252	(13)
(14)	Wind	WSAvg	2.9	2.9	2.5	2.3	2.3	2.8	2.9	3.4	3.4	3.4	3.3	3.1	2.9	(14)
(15)	Precipitation	PrecAvg	36.2	2.7	3.3	2.5	3.1	2.6	4.1	2.8	2.8	2.6	3.2	3.5	3.0	(15)
(16)		PrecMax	45.6	6.1	8.5	9.2	10.9	7.7	12.8	6.0	5.3	5.9	7.9	7.3	6.7	(16)
(17)		PrecMin	25.6	0.7	0.6	0.2	0.3	0.8	0.4	0.4	1.1	0.8	0.8	0.8	0.6	(17)
(18)		PrecStd	6.1	1.5	2.0	2.0	2.5	1.5	3.3	1.9	1.3	1.3	1.6	1.7	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.6	92.9	88.2	78.8	68.9	61.8	61.2	66.4	74.2	82.2	88.5	93.4	(19)
(20)		MCWB	67.9	66.7	65.2	61.0	55.1	52.1	48.6	52.3	55.2	58.9	63.0	65.0	(20)	
(21)		2%	DB	90.9	87.1	82.4	72.7	65.1	58.5	57.4	62.0	69.2	76.5	82.4	86.8	(21)
(22)		MCWB	66.7	65.1	62.9	59.0	53.8	49.4	47.4	49.9	53.6	57.6	61.2	63.6	(22)	
(23)		5%	DB	85.9	82.6	77.7	69.2	62.3	56.2	54.7	58.5	65.6	72.2	77.5	81.8	(23)
(24)		MCWB	65.6	64.6	62.2	57.3	52.5	48.1	46.2	48.2	52.5	56.2	60.8	63.4	(24)	
(25)		10%	DB	81.3	78.9	73.7	66.0	59.6	53.9	52.4	55.7	61.9	68.0	73.2	77.1	(25)
(26)		MCWB	64.5	64.5	61.6	56.0	51.1	46.7	44.8	46.7	51.0	54.9	59.6	62.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.0	70.8	68.7	63.1	58.2	54.0	51.2	54.2	58.1	62.5	67.0	69.3	(27)
(28)		MADB	85.8	82.1	79.6	74.2	63.7	58.8	57.5	62.9	69.3	73.3	78.6	81.6	(28)	
(29)		2%	WB	69.3	68.7	66.2	60.5	55.7	51.3	49.0	51.2	55.6	59.9	64.1	66.8	(29)
(30)		MADB	84.4	80.3	76.0	69.3	62.3	56.8	54.8	59.0	65.8	70.9	76.4	80.0	(30)	
(31)		5%	WB	67.5	67.0	64.2	58.5	53.9	49.4	47.4	49.3	53.6	57.9	62.1	64.7	(31)
(32)		MADB	81.1	78.7	73.7	66.4	60.3	54.1	52.5	56.8	63.0	68.9	73.6	77.6	(32)	
(33)		10%	WB	65.7	65.1	62.4	56.9	52.2	47.9	46.0	47.5	51.7	55.9	60.2	62.8	(33)
(34)		MADB	78.2	75.8	71.8	64.1	58.4	52.4	50.8	54.3	60.4	66.2	71.4	74.8	(34)	
(35)	Mean Daily Temperature Range		MDBR	20.3	18.6	16.1	13.6	10.3	9.0	9.6	11.8	15.0	17.7	18.9	19.9	(35)
(36)		5% DB	MADB	27.9	25.5	22.3	18.2	13.5	12.2	12.9	15.9	20.8	24.4	26.0	27.3	(36)
(37)			MCWBR	12.0	11.1	10.5	9.6	8.7	8.4	8.7	9.7	11.0	11.5	12.1	12.2	(37)
(38)			5% WB	MADB	24.6	21.9	19.1	16.1	12.4	11.0	11.5	15.0	18.6	21.5	23.1	23.6
(39)		MCWBR		12.5	11.4	10.5	9.4	8.5	8.1	8.3	9.7	10.9	11.8	12.3	12.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.392	0.357	0.320	0.306	0.288	0.278	0.270	0.280	0.302	0.319	0.328	0.341	(40)	
(41)		taud	2.350	2.488	2.589	2.604	2.624	2.653	2.687	2.643	2.572	2.543	2.550	2.517	(41)	
(42)		Ebn at Noon	298	302	302	287	275	270	281	293	303	311	317	316	(42)	
(43)		Edh at Noon	42	35	29	25	21	19	20	23	29	33	34	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2080	1784	1500	1114	811	637	736	1005	1383	1735	1945	2129	(44)	
(45)		RadStd	148	134	116	85	41	46	49	59	98	105	143	152	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.92	+0.63	N/A	N/A	+1.17	+1.12	N/A	-194	+310	+122	(46)
(47)	Regional (25 neighbors)	+0.63	N/A	N/A	N/A	N/A	N/A	N/A	-157	+227	+85	(47)

Nomenclature: See separate page